Model Documentation: Governing Equations and Economics for a Solar-Driven, Single-Bed PAM–LiCl SAWH Device

Reference and scope

This document describes the lumped device + techno-economic model implemented in `solar_lumped`, following Wilson & Díaz-Marín *Device* (2025) Eqs. 1–6 and Note S1 (COMSOL Table S3 parameters) for the physics, and the package's own `economics/` module for cost. The device is a single hydrogel bed that alternates between open *absorption* (night) and sealed *desorption* (day) half-cycles. This is the default (`desorption_solver="quasi_steady"`) configuration; one alternate exists in code but is out of scope here: `segregated` / `coupled_bdf` desorption integrators, which trade the quasi-steady algebraic thermal solve for COMSOL-mimicking or fully transient alternatives (`device_config.py`).

Base case is Case 2, not Wilson's original paper physics. As of 2026-07, `DeviceConfig.thermal_params()` – the physics every plain `DeviceConfig()` call gets unless it explicitly overrides `thermal=` – defaults to real absorber/glass IR emissivities ($\varepsilon_{abs,ir} = 0.05$, $\varepsilon_{gl,ir} = 0.95$, the "selective surface" variant, Case 2 in `sawh_bayesopt.design_space.CASE_EPS_IR`) rather than Wilson's original blackbody/cavity approximation (Case 1). Every equation, assumption, and default value below describes Case 2 unless stated otherwise. Case 1 (Wilson's exact paper physics) remains reachable via an explicit `DeviceThermalParams(eps_abs_ir=1.0,eps_glass_ir=1.0)` (or the equivalent `eps_abs_ir=eps_glass_ir=None`, still numerically identical – see Section 3) override, and is exactly what the paper-reproduction scripts under `analysis/paper_recreation/wilson/` use – those scripts build their own `DeviceThermalParams` directly rather than going through `DeviceConfig.thermal_params()`, so they are unaffected by this default and continue to reproduce Wilson's digitized figures under Case 1. Case 3 (idealized optical-material limits, $\varepsilon_{abs,ir} = \varepsilon_{gl,ir} = 0.0$) is a third explicit sensitivity variant, also not the default.

Subscripts: g = hydrogel; abs = solar absorber; gl = cover glass; $cond$ = aluminum condenser; amb = ambient.

How to use the Source columns below: cells already filled cite the in-repo file or paper section the value/range comes from. Cells marked **TBD** have no literature or vendor citation on file yet – the value is either an as-implemented constant or an arbitrarily chosen sweep bound with no documented justification. Fill these in as sources are found; do not treat a TBD as validated.

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1 Model design

1.1 Device overview

A footprint-area device stacks (top to bottom, day-side up): cover glass → air/insulation gap → solar absorber (aluminum) → hydrogel-salt composite bed → vapor gap → aluminum condenser (finned, ambient-cooled). All balances are per unit footprint area (m^2) unless noted. During **desorption** (daytime), the chamber is sealed: solar radiation heats the absorber, which conducts heat into the gel; the gel desorbs water vapor across the vapor gap to the cooler condenser, where it condenses and is collected. During **absorption** (nighttime), the chamber is opened to ambient air; the gel re-hydrates by uptake from ambient humidity, with no solar input and no condenser dynamics. An optional uncovered variant (`has_glass=False`) removes the glass/insulation-gap stack entirely and radiates the absorber directly to a sky/ambient sink.

1.2 State variables

Symbol	Units	Role
c_w	$\mathrm{mol\,m^{-3}}$	Brine water concentration in gel (Eq. 5 state)
H	m	Hydrogel thickness (Eq. 6 state; reference H_0)
T_{cond}	°C	Condenser temperature (transient during desorption only)
T_g, T_{abs}, T_{gl}	°C	Quasi-steady thermal states (algebraic each step)

Absorption phase ODEs: $\dot{c}_w$, $\dot{H}$ with $Q_{solar} = 0$, $\dot{m}_{des} = 0$, $\dot{T}_{cond} = 0$. **Desorption phase ODEs:** $\dot{c}_w$, $\dot{H}$, $\dot{T}_{cond}$ with T_g, T_{abs}, T_{gl} from the coupled thermal solve.

1.3 Daily cycle structure

One simulated day is one absorption half-cycle followed by one desorption half-cycle, gel state carried forward between them. The absorption/desorption split follows the forcing weather profile's own real day/night boundary (solar-radiation threshold crossing, so the two halves are not forced to equal length) in `real`/replay weather modes; the `baseline` synthetic-profile mode instead defaults to a fixed 12 h/12 h split (`weather/profiles.py::PHASE_HOURS`) so that single-axis parameter sweeps hold the forcing schedule constant. Multi-day runs either warm up for a fixed cycle count or use Aitken Δ^2 extrapolation (`find_cyclic_state`) to locate the true steady periodic post-desorption (c_w, H) state before reporting a single representative day.

2 Physics model assumptions

- **Lumped / 0-D:** the gel, absorber, glass, and condenser are each a single uniform temperature node per footprint area; no through-thickness gradients are resolved.
- **Quasi-steady thermal solve during desorption:** T_g, T_{abs}, T_{gl} are found by root-finding the steady residual form of Eqs. 1, 3, 4 at every ODE evaluation (assumes the thermal response is fast relative to the hydration/mass-transfer timescale). Only T_{cond} carries real thermal inertia (finite aluminum slab thermal mass) during desorption.
- **Absorption thermal shortcut:** during absorption, $T_g = T_{abs} = T_{gl} = T_{amb}$ is assumed outright (fast gel thermal storage, `coupled_dynamics.py` Note S1 Eq. S1) rather than solved from Eqs. 1/3/4 – there is no solar input and no condenser coupling to balance against.
- **Radiative exchange (base case, Case 2 "selective surface"):** absorber↔glass and glass↔ambient exchange use real IR emissivities $\varepsilon_{abs,ir} = 0.05$, $\varepsilon_{gl,ir} = 0.95$ (Table 1), not Wilson's typeset Eqs. 3–4 blackbody/cavity approximation ($\varepsilon = 1$). Setting both to 1.0 (Case 1) reproduces Wilson's original paper physics exactly; setting both to 0.0 (Case 3) gives the idealized optical-material-limits sensitivity variant. See Section 3.
- **Radiative sink temperature:** the default sink for the outermost surface is ambient air temperature; an optional `sky_temp_depression_c` lowers the effective radiant sky temperature below air temperature, needed to reproduce the COMSOL Atacama field-test curves (surface-to-ambient radiation with a real sky, not blackbody air).
- **Water inventory basis:** c_w and H are tracked as separate states, but the sorbate mass used for activity/uptake calculations is referenced to the fixed fabrication thickness H_0, not the swollen $H(t)$ – swelling only feeds back into vapor-gap convection $(L_g - H)$ and gel–absorber conductance (H/k_{gel}). This avoids double-counting dilution (see `salt_properties.py::pam_licl_gravimetric_uptake_g_g`).
- **Water activity (LiCl):** brine activity from the Conde (2004) correlation, capped during absorption only by the measured PAM–LiCl DVS isotherm, i.e. $a_w = \max(a_{w,brine}, a_{w,DVS})$ in absorption; brine activity alone in desorption. Non-LiCl salts (NaCl, $CaCl_2$, $MgCl_2$) use brine activity only, with no DVS cap (no isotherm digitized for them) – out of this document's primary scope but present in code.
- **Mass-transfer coefficient:** absorption uses a fixed empirical open-chamber convection coefficient $g_{chamber}$; desorption derives g from the Hollands natural-convection $h_{conv,g}$ via a Lewis-number-unity heat/mass analogy (Note S1 Eq. S5). No forced convection is modeled in either phase.
- **Self-consistent desorption coupling:** $\dot{m}_{des}$ is found as the root of $\dot{m}_{calc}(\dot{m}_{des}) - \dot{m}_{des} = 0$ (Brent's method) so that the gel heat balance (Eq. 1) and the mass-transfer equations (Eqs. 5–6) agree on the same rate at every step.

- **Scope:** default salt is LiCl with a hydrogel sorbent. A MOF-sorbent path and three additional deliquescent salts exist in code with their own property tables but are not documented further here.

- **Initial condition:** c_w initializes at equilibrium with the ~20 % RH ambient the gel is fabricated at (DVS isotherm, LiCl), unless a cyclic steady-state or explicit override is requested instead.

3 Thermal balances (Wilson Eqs. 1, 3, 4)

During each integration step the gel, absorber, and glass temperatures satisfy the steady residual form (Eqs. 1, 3, 4 set to zero):

$$U_{gel}(T_{abs} - T_g) = h_{conv,g}(T_g - T_{cond}) + \dot{m}_{des}\, h_{des} + \varepsilon_{gc}\sigma(T_g^4 - T_{cond}^4) \tag{1}$$

$$k_{air}\frac{T_{abs} - T_{gl}}{L_{ins}} + \varepsilon_{ag}\sigma(T_{abs}^4 - T_{gl}^4) = h_{amb}(T_{gl} - T_{amb}) + \varepsilon_{ga}\sigma(T_{gl}^4 - T_{sky}^4) \tag{2}$$

$$\varepsilon_{abs}\,\tau_{glass}\,Q_{solar} = k_{air}\frac{T_{abs} - T_{gl}}{L_{ins}} + \varepsilon_{ag}\sigma(T_{abs}^4 - T_{gl}^4) + U_{gel}(T_{abs} - T_g) \tag{3}$$

where $T_{sky} = T_{amb} - \Delta T_{sky}$ (default $\Delta T_{sky} = 0$). In the uncovered variant (`has_glass=False`), Eqs. (2)–(3) collapse to a single absorber-to-sky balance $\varepsilon_{abs}Q_{solar} = h_{amb}(T_{abs} - T_{amb}) + \varepsilon_{abs}\sigma(T_{abs}^4 - T_{sky}^4) + U_{gel}(T_{abs} - T_g)$ with $T_{gl} \equiv T_{amb}$.

Correlations (Note S1):

- $h_{conv,g}$: Hollands et al. natural convection between parallel plates in the vapor gap $L_g - H$ (tilt-dependent; Eqs. S3–S4).

- $\varepsilon_{gc} = 1/(1/\varepsilon_{gel} + 1/\varepsilon_{Al} - 1)$: parallel-plate radiative exchange (Eq. S2).

- U_{gel}: series-resistance gel–absorber conductance (updated from the single-layer approximation in earlier drafts of this document to match `table_s3.py::u_gel_w_m2_k`):

$$\frac{1}{U_{gel}} = \frac{L_{Al}}{k_{Al}} + \frac{L_{silicone}}{k_{silicone}} + \frac{H(t)}{k_{gel}} \tag{4}$$

  (aluminum backing + silicone bonding layer + hydrogel conduction in series; H floored at $H_0/4$ to avoid a singular resistance).

- $\varepsilon_{ag}, \varepsilon_{ga}$ (base case, Case 2): $\varepsilon_{ag} = \text{parallel\_plate\_emissivity}(\varepsilon_{abs,ir}, \varepsilon_{gl,ir}) = 1/(1/0.05 + 1/0.95 - 1) \approx 0.0508$ (absorber→glass, same parallel-plate formula as ε_{gc} above); $\varepsilon_{ga} = \varepsilon_{gl,ir} = 0.95$ directly (glass radiates to ambient/sky, not a cavity). Case 1 ($\varepsilon_{abs,ir} = \varepsilon_{gl,ir} = 1.0$) collapses both to 1.0, reproducing the blackbody base model exactly; Case 3 (both 0.0) collapses both to 0.0 (no net IR exchange).

- Absorption: $Q_{solar} = 0$; desorption: $Q_{solar} \geq 0$ from the weather profile (or a fixed baseline value for controlled sweeps).

4 Condenser energy balance (Wilson Eq. 2, transient)

During desorption only:

$$(\rho c_p L)_{cond} \frac{dT_{cond}}{dt} = h_{conv,g}(T_g - T_{cond}) + \dot{m}_{des}\, h_{fg} + \varepsilon_{gc}\sigma(T_g^4 - T_{cond}^4) - A_r\, h_{amb}(T_{cond} - T_{amb}) \quad (5)$$

where A_r is the fin area ratio and $h_{conv,cond} = A_r\, h_{amb}$ (or $A_r\, h_{amb,cond}$ when a separate fan-forced condenser convection coefficient is supplied).

5 Mass transport (Wilson Eqs. 5–6)

$$\frac{dc_w}{dt} = \frac{g}{H_0} \frac{P_{sat}(T_g)}{RT_g} (C_R - a_w) \quad (6)$$

$$\frac{dH}{dt} = g\, \frac{MW_w}{\rho_{sol}} \frac{P_{sat}(T_g)}{RT_g} (C_R - a_w) \quad (7)$$

Wilson's Eq. 6 as printed omits the g prefactor, but the bare product $\frac{MW_w}{\rho_{sol}} \frac{P_{sat}}{RT_g}$ is dimensionless (m^3/mol $\times$ mol/m^3), so dH/dt as printed cannot carry units of m/s. Including g (same mass-transfer coefficient as Eq. (6)) is required both for dimensional consistency and to match the desorption mass-rate identity $\dot{m}_{des} = -\rho_{sol} A_c\, dH/dt$ used elsewhere in the paper against the flux-based $\dot{m}_{des} = MW_w\, g\, \frac{P_{sat}}{RT_g}(C_R - a_w)$. This implementation therefore keeps the g factor.

Concentration ratio C_R:

$$C_R = RH_{amb} \quad \text{(absorption, open chamber)} \quad (8)$$

$$C_R = \frac{P_{sat}(T_{cond})}{P_{sat}(T_g)} \frac{T_g + 273.15}{T_{cond} + 273.15} \quad \text{(desorption, vapor gap)} \quad (9)$$

Mass-transfer coefficient g:

$$g = g_{chamber} \quad \text{(absorption)} \quad (10)$$

$$g = h_{conv,g} \frac{D_{air}}{k_{air}} \quad \text{(desorption; Lewis-number analogy, Note S1 Eq. S5)} \quad (11)$$

Water activity a_w (PAM–LiCl): brine activity from salt thermodynamics, capped by the measured DVS isotherm during absorption (Note S2). During desorption, brine activity alone.

Saturation vapor pressure $P_{sat}(T)$: Conde (2004) Saul–Wagner correlation, with a Tetens-equation fallback if the primary correlation is non-finite (`salt_properties.py::saturation_vapor_pressure_pa`).

Desorption mass flux from gel inventory:

$$\dot{m}_{des} = MW_w \left(-\dot{c}_w\, H - c_w\, \dot{H}\right), \qquad \dot{m}_{des} \geq 0 \quad (12)$$

6 Thermal–mass coupling constraint

During desorption, $\dot{m}_{des}$ in Eq. (1) must equal the flux implied by Eqs. (6)–(12). The implementation finds $\dot{m}_{des}$ as the root of $f(\dot{m}_{des}) = \dot{m}_{calc}(\dot{m}_{des}) - \dot{m}_{des} = 0$ (Brent's method), ensuring Eqs. 1 and 5–6 are self-consistent at each ODE evaluation.

7 Constraints and cycle-level outputs

- $c_w \in [c_{w,\min}, c_{w,\max}]$; rates clipped at bounds.
- $H \geq H_0$; swelling allowed in absorption ($\dot{H} \geq 0$ at $H = H_0$); no shrinkage below H_0 in desorption.
- Desorption: enforce $\dot{c}_w \leq 0$, $\dot{H} \leq 0$.
- Vapor gap below 7 mm (Wilson's thermobuoyancy/transport floor, $L_g - H$) is treated as mass-transfer-inhibited: $g \to 0$ rather than extrapolating the Hollands correlation outside its validated range.
- Initial c_w from fabrication equilibrium RH (DVS isotherm at H_0) unless overridden.
- Daily water yield is the condenser-collected mass over one desorption phase; thermal efficiency $\eta_{th} = \dot{m}_{water}\, h_{fg} / \int Q_{solar}\, dt$, integrated over the desorption phase only (blending in the near-zero-solar absorption phase would dilute the ratio without adding real signal).

8 Physics model parameters

Table 1 lists parameters actively exposed to parameter sweeps / Bayesian optimization, with the ranges the codebase already uses. Table 2 lists fixed material and correlation constants with no documented sweep range.

Table 1: Swept / design physics parameters and their documented ranges.

Parameter	Symbol	Baseline	Sweep range	Source
Hydrogel reference thickness	H_0	4 mm	1.0 mm to 10.0 mm	`table_s3.H0_M`; range: `parameter_sweep.py` / `lcow_economic_params.csv` (`hydrogel_thickness_min/max_m`)
Vapor gap	L_g	40 mm	7.0 mm to 60.0 mm	`table_s3.L_G_M`; range: `parameter_sweep.py` / `design_space.py` `DesignBounds`; lower bound = `table_s3.VAPOR_GAP_TRANSPORT_MIN_M`
Insulation gap	L_{ins}	5 mm	1.0 mm to 20.0 mm	`table_s3.L_INS_M`; range: `parameter_sweep.py` / `design_space.py`
Condenser fin area ratio	A_r	7.1	3.0 to 12.0	`table_s3.FIN_AREA_RATIO`; range: `parameter_sweep.py` / `grid_param_sweep.py` / `design_space.py`

Parameter	**Symbol**	**Baseline**	**Sweep range**	**Source**
Tilt angle	θ	30° (Fig. 2 baseline; 35° used as the Sherlock grid-sweep baseline)	0° to 60°	`table_s3.TILT_DEG`; range: `parameter_sweep.py` / `design_space.py`
Salt:polymer ratio	S/L	4.0	1.0 to 8.0	`table_s3` default / `DeviceConfig`; range: `parameter_sweep.py` / `design_space.py`
Absorber emissivity	ε_{abs}	0.95	0.85 to 0.95 (3 discrete levels)	`table_s3.EPS_ABS`; range: `sherlock_param_sweep.tex` / `grid_param_sweep.py--eps-abs`
Glass transmittance	τ_{glass}	0.90	0.80 to 0.90 (3 discrete levels)	`table_s3.TAU_GLASS`; range: `sherlock_param_sweep.tex` / `grid_param_sweep.py--tau-glass`
Desorption enthalpy (LiCl)	h_{des}	$2.32 \times 10^6\,\mathrm{J\,kg^{-1}}$ (Table S3 COMSOL value)	$1.8 \times 10^6\,\mathrm{J\,kg^{-1}}$ to $3.2 \times 10^6\,\mathrm{J\,kg^{-1}}$	`table_s3.H_DES_J_PER_KG`; range: `parameter_sweep.py`; cf. literature range in `salt_heat_of_desorption.csv` (LiCl $\approx 2.85 \times 10^6\,\mathrm{J\,kg^{-1}}$, Díaz-Marín)
Salt weight factor (DVS-cap MW scaling, dimensionless)	—	1.0	0.826 to 1.180 (i.e. effective LiCl formula weight $35\,\mathrm{g\,mol^{-1}}$ to $50\,\mathrm{g\,mol^{-1}}$)	`parameter_sweep.py::make_sweep_params`
Ambient convection coefficient (baseline profile)	h_{amb}	$10.0\,\mathrm{W\,m^{-2}\,K^{-1}}$	$1.0\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $12.5\,\mathrm{W\,m^{-2}\,K^{-1}}$	`weather/profiles.py::FIXED_H_AMB_W_M2_K`; range: `parameter_sweep.py`
Uptake RH / ambient relative humidity (baseline profile)	RH_{amb}	0.5	0.15 to 0.80	`parameter_sweep.py` (`humidity_high`; baseline-mode profile only)
Solar irradiance (baseline profile)	Q_{solar}	$600\,\mathrm{W\,m^{-2}}$	$400\,\mathrm{W\,m^{-2}}$ to $800\,\mathrm{W\,m^{-2}}$	`parameter_sweep.py` (`solar_irradiance_w_per_m2`; baseline-mode profile only – real-weather mode uses fetched irradiance instead)
Ambient temperature (baseline profile)	T_{amb}	$25\,°\mathrm{C}$	$10\,°\mathrm{C}$ to $40\,°\mathrm{C}$	`parameter_sweep.py` (`temperature_c`; baseline-mode profile only)

Parameter	Symbol	Baseline	Sweep range	Source
Absorber/glass real IR emissivity pair (base case; Case 2, "selective surface")	$\varepsilon_{abs,ir}, \varepsilon_{gl,ir}$	(0.05, 0.95) – `DeviceConfig.thermal_params()` default as of 2026-07	categorical, not a continuous sweep: `case1` = (1.0, 1.0) reproduces Wilson's blackbody physics exactly; `case3` = (0.0, 0.0) idealized optical limits	`device_config.py::DeviceConfig.thermal_params()`; case registry: `sawh_bayesopt.design_space.py::CASE_EPS_IR` – **TBD** literature basis for the case2/case3 numeric pairs

Table 2: Fixed physics material / correlation constants (no documented sweep range).

Parameter	Symbol	Value	Source
Chamber convection (absorption)	$g_{chamber}$	$0.0085\,\mathrm{m\,s^{-1}}$	`table_s3.G_CHAMBER_M_S`; duplicated in economics CSV as `mass_transfer_convection_coefficient_m_s` – **TBD** calibration source
Solution/brine density	ρ_{sol}	$1250.2\,\mathrm{kg\,m^{-3}}$ (LiCl); salt-dependent	`salt_catalog.csv` ($NaCl/CaCl_2/MgCl_2$ $\approx 1200\,\mathrm{kg\,m^{-3}}$)
Condensation enthalpy	h_{fg}	$2.256 \times 10^6\,\mathrm{J\,kg^{-1}}$	`table_s3.H_FG_J_PER_KG` – **TBD** source (steam-table value assumed)
Gel emissivity	ε_{gel}	1.0	`table_s3.EPS_GEL` – **TBD**
Condenser (Al) emissivity	ε_{Al}	0.05	`table_s3.EPS_AL` – **TBD**
Sky temperature depression	ΔT_{sky}	0 K (no depression; non-zero used only to calibrate the COMSOL Atacama field-test curves)	`DeviceThermalParams.sky_temp_depression_c` – **TBD** calibrated value / general range
Air thermal conductivity	k_{air}	$0.0286\,\mathrm{W\,m^{-1}\,K^{-1}}$	`table_s3.K_AIR_W_M_K`
Aluminum thermal conductivity	k_{Al}	$167.0\,\mathrm{W\,m^{-1}\,K^{-1}}$	`table_s3.K_AL_W_M_K` (Table S3)
Hydrogel thermal conductivity	k_{gel}	$0.6\,\mathrm{W\,m^{-1}\,K^{-1}}$	`table_s3.K_GEL_W_M_K` (Table S3)
Cover glass thermal conductivity	k_{glass}	$1.2\,\mathrm{W\,m^{-1}\,K^{-1}}$	`table_s3.K_GLASS_W_M_K` – **TBD**
Silicone bonding-layer conductivity	$k_{silicone}$	$0.2\,\mathrm{W\,m^{-1}\,K^{-1}}$	`table_s3.K_SILICONE_W_M_K` – **TBD**
Aluminum density / specific heat	$\rho_{Al}, c_{p,Al}$	$2700\,\mathrm{kg\,m^{-3}}$, $900\,\mathrm{J\,kg^{-1}\,K^{-1}}$	`table_s3.RHO_AL_KG_M3` / `CP_AL_J_KG_K` – standard aluminum handbook values, **TBD** exact citation
Cover-glass density / specific heat	$\rho_{glass}, c_{p,glass}$	$2230\,\mathrm{kg\,m^{-3}}$, $830\,\mathrm{J\,kg^{-1}\,K^{-1}}$	`table_s3.RHO_GLASS_KG_M3` / `CP_GLASS_J_KG_K` (borosilicate) – **TBD**

Parameter	Symbol	Value	Source
Gel specific heat	$c_{p,gel}$	$3500\,\mathrm{J\,kg^{-1}\,K^{-1}}$	`table_s3.CP_GEL_J_KG_K` (hydrated PAM–LiCl, water-dominated) – **TBD**
Water-vapor-in-air diffusivity	D_{air}	$2.62 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$	`correlations.py::D_AIR_M2_S` (Note S1 Sh = Nu analogy) – **TBD** exact source
Air kinematic viscosity / Prandtl number	ν_{air}, Pr_{air}	$1.5 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$, 0.71	`correlations.py` – standard air properties near 25 °C, **TBD** citation
Gravitational acceleration	g	$9.81\,\mathrm{m\,s^{-2}}$	`correlations.py::GRAVITY_M_S2`
Stefan–Boltzmann constant	σ	$5.670\,374\,419 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$	CODATA
Vapor-gap transport floor	—	7 mm	`table_s3.VAPOR_GAP_TRANSPORT_MIN_M` (Wilson §2.2 thermobuoyancy/transport floor)

9 Economics model assumptions

- **LCOW** is a standard fixed-charge-rate techno-economic calculation: annualized cost divided by annual usable water volume. CAPEX is amortized via a capital recovery factor (CRF) at a fixed discount rate over a fixed device lifetime – not a full multi-year cash-flow model with price escalation.
- **NPV / payback** (`npv.py`) reuses the same cost components as LCOW, but restructures CAPEX as an upfront year-0 payment against a level (non-escalating) annual net cash flow (revenue at an assumed water price, minus OPEX), discounted at the same rate – a level-annuity NPV, not a full pro forma.
- **CAPEX** is a fixed per-footprint bill of materials (BOM), summed from six line items, scaled uniformly by a `total_investment_factor` (contingency / installation multiplier that is easy to sweep even though the underlying line items themselves are not).
- **Maintenance** is a fixed fraction of CAPEX per year – not tied to any modeled failure or service-call process.
- **Sorbent replacement:** the hydrogel is replaced on a fixed interval (`hydrogel_lifetime_years`); its replacement cost is the dry composite mass per footprint area times a blended cost (catalog salt price + fixed acrylamide/additive costs, Table S1). MOF sorbent instead uses mass × price / lifetime directly.
- **Utilization factor** is a single multiplicative derating of the physics-simulated gross annual yield (daily yield × 365 × cycles/day), meant to stand in for real-world downtime / non-ideal weather years – not a modeled reliability/availability process.
- **Electricity cost** only applies to the optional active/electric-heat-assist mode; the reproduced passive-solar baseline has zero electric heat and therefore zero electricity cost.
- **Out of scope:** water transport/distribution, land, installation labor, permitting, and end-of-life salvage value are not modeled.

- **No escalation:** all costs are unescalated real USD (no inflation or fuel/material price growth curves).
- **Water price** (NPV/payback only) is an external scenario input, not derived from any market model.

10 Economics governing equations

Capital recovery factor:

$$CRF = \frac{i(1+i)^L}{(1+i)^L - 1} \tag{13}$$

where i is the discount rate and L the device lifetime (years).

Sorbent (hydrogel) replacement, annualized:

$$C_{sorbent} = \frac{c_{gel}\, m_{dry}}{L_{gel}}, \qquad c_{gel} = \frac{c_{salt}\,(S/L) + c_{acrylamide}}{1 + S/L} + c_{additives} \tag{14}$$

where $m_{dry} = H_0\, \rho_{gel}$ is the dry composite mass per footprint area, S/L is the salt:polymer ratio, and L_{gel} is the hydrogel replacement interval (years). (MOF sorbent: $C_{sorbent} = m_{MOF}\, c_{MOF}/L_{gel}$ directly.)

Annualized device cost:

$$C_{annual} = CRF \cdot f_{inv}\, C_{device} + C_{sorbent} + f_{maint}\, f_{inv}\, C_{device} + C_{energy,fixed} + C_{elec} + C_{cycle,extra} \tag{15}$$

where $C_{device} = \sum_j C_{BOM,j}$ (six BOM line items, Table 4), f_{inv} is the total investment factor, f_{maint} the maintenance cost fraction, $C_{elec} = p_{elec}\, \dot{Q}_{elec}\, t_{des} \cdot 365/1000$ is the optional active-heating electricity cost (USD/kWh $\times$ W/m^2 $\times$ h/day $\times$ days/yr, converted to kWh), and $C_{cycle,extra}$ charges a fixed energy cost per half-cycle beyond one per day.

LCOW:

$$LCOW = \frac{C_{annual}}{f_{util}\left(\dfrac{365 \cdot n_{cyc} \cdot Y_{daily}}{\rho_w}\right)} \tag{16}$$

where Y_{daily} is the simulated daily water yield (kg/m^2), n_{cyc} the number of half-cycles per day, ρ_w the density of water (kg/m^3), and f_{util} the utilization factor. The cost breakdown (`lcow_cost_breakdown_from_daily_yield`) reports each term of Eq. (15) individually, divided by the same denominator, so CAPEX / sorbent / maintenance / energy segments sum exactly to $LCOW$.

NPV and payback:

$$NPV = -f_{inv}\, C_{device} + (R_{annual} - OPEX_{annual}) \cdot \frac{1 - (1+i)^{-L}}{i} \tag{17}$$

$$R_{annual} = f_{util}\left(\frac{365 \cdot n_{cyc} \cdot Y_{daily}}{\rho_w}\right) p_{water} \tag{18}$$

where p_{water} is the assumed water sale price (USD/m^3) and $OPEX_{annual}$ is Eq. (15) without the CAPEX/CRF term. Simple payback is $CAPEX/(R_{annual} - OPEX_{annual})$; discounted payback solves $1 - (1 - i\, CAPEX/(R_{annual} - OPEX_{annual})) = (1+i)^{-t}$ for t (`npv.py::npv_from_daily_yield`).

11 Economics model parameters

Table 3 lists economics parameters exposed to sweeps (`parameter_sweep.py::make_sweep_params`); Table 4 lists fixed inputs (BOM line items, per-salt prices, and scenario constants) with no documented sweep range.

Table 3: Swept economics parameters and their documented ranges.

Parameter	Symbol	Baseline	Sweep range	Source
Discount rate	i	0.08	0.04 to 0.12	`lcow_economic_params.csv`; range: `parameter_sweep.py`
Device lifetime	L	20 yr	10 yr to 30 yr (integer)	`lcow_economic_params.csv`; range: `parameter_sweep.py`
Utilization factor	f_{util}	0.9	0.7 to 1.0	`lcow_economic_params.csv`; range: `parameter_sweep.py`
Total investment factor	f_{inv}	1.0	0.7 to 1.5	`lcow_economic_params.csv`; range: `parameter_sweep.py`
Maintenance cost fraction	f_{maint}	0.05 yr^{-1}	0.02 to 0.10 yr^{-1}	`lcow_economic_params.csv`; range: `parameter_sweep.py`
Electricity price	p_{elec}	0.10 USD kW^{-1} h	0.05 USD kW^{-1} h to 0.30 USD kW^{-1} h	`lcow_economic_params.csv`; range: `parameter_sweep.py`
Acrylamide price	$c_{acrylamide}$	1.10 USD kg^{-1}	0.55 USD kg^{-1} to 1.65 USD kg^{-1} (±50% of baseline)	`lcow_economic_params.csv` (Table S1 note); range: `parameter_sweep.py`
Additives price (composite)	$c_{additives}$	0.007 214 8 USD kg^{-1}	0.003 607 4 USD kg^{-1} to 0.010 822 2 USD kg^{-1} (±50% of baseline)	`lcow_economic_params.csv` (APS, MBAA, TEMED per Table S1); range: `parameter_sweep.py`
Hydrogel lifetime	L_{gel}	1.0 yr	0.5 yr to 2.0 yr	`lcow_economic_params.csv`; range: `parameter_sweep.py`
Water price (NPV/payback scenario input)	p_{water}	5.0 USD m^{-3}	1.0 USD m^{-3} to 25.0 USD m^{-3}	`parameter_sweep.py` (`_BASELINE_WATER_PRICE_USD_PER_M3`) – **TBD** market basis
Salt:polymer ratio (dual physics/cost role – see Table 1)	S/L	4.0	1.0 to 8.0	See Table 1; also sets $C_{sorbent}$ via Eq. (14)

Table 4: Fixed economics parameters and BOM line items (no documented sweep range).

Parameter	Symbol	Value	Source
BOM: aluminum heater	$C_{BOM,1}$	11.382 USD m^{-2}	`lcow_economic_params.csv` (`device_bom_aluminum_heater`) – **TBD** itemized cost basis

Parameter	Symbol	Value	Source
BOM: condenser	$C_{BOM,2}$	28.455 USD m^{-2}	`lcow_economic_params.csv` (`device_bom_condenser`) – **TBD**
BOM: acrylic enclosure	$C_{BOM,3}$	5.95 USD m^{-2}	`lcow_economic_params.csv` (`device_bom_acrylic_enclosure`) – **TBD**
BOM: insulation	$C_{BOM,4}$	2.0 USD m^{-2}	`lcow_economic_params.csv` (`device_bom_insulation`) – **TBD**
BOM: clamps	$C_{BOM,5}$	3.474 USD m^{-2}	`lcow_economic_params.csv` (`device_bom_clamps`) – **TBD**
BOM: fixtures	$C_{BOM,6}$	1.0 USD m^{-2}	`lcow_economic_params.csv` (`device_bom_fixtures`) – **TBD**
Total device BOM	C_{device}	52.261 USD m^{-2} (sum of the above)	derived
Salt price (per catalog)	c_{salt}	LiCl 0.55; NaCl 0.045; $CaCl_2$ 0.17; $MgCl_2$ 0.019 85 (USD kg^{-1})	`salt_catalog.csv` – **TBD** vendor/date basis
Fixed annual energy cost	$C_{energy,fixed}$	0.0 USD yr^{-1}	`lcow_economic_params.csv` (`energy_cost_usd_per_year`) – no sweep documented
Extra half-cycle energy cost	—	0.0 USD d^{-1}	`lcow_economic_params.csv` (`energy_cost_usd_per_extra_half_cycle_per_day`) – no sweep documented
Desorption hours per day	t_{des}	12.0 h	`lcow_economic_params.csv` (`desorption_hours_per_day`); tie to the physics day/night split (`weather/profiles.py::PHASE_HOURS`) is manual, not automatic – keep the two consistent when sweeping cycle length
Max electric heat (optimization bound, not baseline)	$\dot{Q}_{elec,max}$	1500.0 W m^{-2}	`lcow_economic_params.csv` (`max_electric_heat_w_per_m2`) – an optimizer bound, not a swept axis; baseline passive device has $\dot{Q}_{elec} = 0$
Include desorption enthalpy in T_g solve	—	`true` (boolean, not swept)	`lcow_economic_params.csv` (`include_desorption_enthalpy`)
Water density	ρ_w	1000 kg m^{-3}	`lcow_economic_params.csv` (`water_density_kg_per_l` × `l_per_m3`)

12 Integration

The coupled physics system is integrated with SciPy `solve_ivp` (Radau, stiff). State during desorption: $\mathbf{y} = [c_w, H, T_{cond}]$; during absorption: $\mathbf{y} = [c_w, H]$. LCOW/NPV are evaluated post-hoc from the resulting daily yield – they do not feed back into the ODE integration.

13 Open items

- Every **TBD** cell above is an as-implemented constant or sweep bound with no literature/vendor citation currently attached in the repository – filling these in (or replacing the value where a better source disagrees) is the natural next step for anyone extending this document.

- The physics/economics duplication of the mass-transfer convection coefficient ($g_{chamber}$ in `table_s3.py` vs. `mass_transfer_convection_coefficient_m_s` in `lcow_economic_params.csv`) is a single physical quantity kept in two files; a change to one without the other would silently desynchronize the model.

- `desorption_hours_per_day` (economics, fixed at 12h) is not derived from the physics day/night split, which can vary by site/season in real-weather mode – a mismatch here biases the electricity-cost term for the optional active-heating case.

- **Case 2 default (2026-07):** `sawh_bayesopt` (`design_space.py`, `evaluator.py`, `bayesopt.py`, and the `run_bayesopt.py`/`hp_sweep.py` CLI flags) also now defaults its own case-selection axis to `case2`, and `case1` there is now built with an explicit `eps_abs_ir=eps_glass_ir=1.0` override rather than relying on omission – numerically identical to the old `None,None` encoding (verified; $\varepsilon_{ag} = \varepsilon_{ga} = 1.0$ either way), so historical `case1 cache.jsonl` entries remain valid, but every `to_device_config_kwargs` call now always includes a `"thermal"` key (previously omitted for `case1`), which changes that request's cache key. `scripts/grid_param_sweep.py`'s `--eps-abs-ir`/`--eps-glass-ir` CLI flags (used for the already-executed Sherlock global-grid sweep, `sherlock_param_sweep.tex`) were deliberately *not* changed – they still default to `None` (Case 1) to keep that documented, already-run production sweep's baseline meaning intact; a rerun under Case 2 would need those flags set explicitly.